1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 4 · Let's start simple…What is the equation of this line?And more importantly, why is it that?



Let's start simple...

What is the equation of this line?

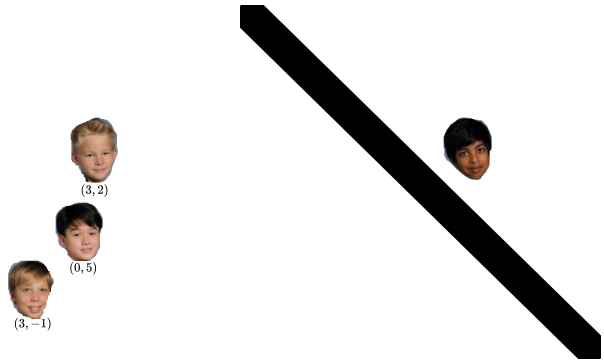
And more importantly, why is it that?

4
3
2
1
-1
-2
-3
-4

x -5 -4 -3 -2 -1 0 1 2 3 4 5 6

For any point we pick on the line, the x value is always 2.

Chapter 5 · Original slide 5 · Lines and Equations of Lines



! A line consists of all points which satisfy some equation in terms of x and/or y.

Anirudh's Line Club

Membership Rules:

$2x + y = 5$

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 6 · Examples

This means we can substitute the values of a coordinate into our equation whenever we know the point lies on the line.

Find the coordinate of the point where the line $2x + y = 5$ cuts the x-axis.

The point $(5, a)$ lies on the line with equation $y = 3x + 2$. Determine the value of a.

Chapter 5 · Original slide 7 · Test Your Understanding

Determine where the line $x + 2y = 3$ crosses the:

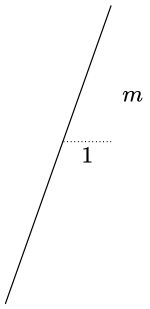
a) y-axis:

b) x-axis:

What mistakes do you think it's easy to make?

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 8 · Recap of gradient



The steepness of a line is known as the gradient.

It tells us what y changes by as x increases by 1.

So if the y value increased by 6 as the x value increased by 2, what is y increasing by for each unit increase of x?

How would that give us a suitable formula for the gradient m?

$m = \frac{\Delta y}{\Delta x}$

Δ is the (capital) Greek letter "delta" and means "change in".

Textbook Note:

The textbook uses $m = \frac{y_2 - y_1}{x_2 - x_1}$ for two points (x_1, y_1) and (x_2, y_2).

Reasons I don't use it for non-algebraic coordinates:

• Students often get the y_1 and y_2 the wrong way round (or with the x's)

• Students often make sign errors when dealing with negatives, e.g. $(-3) - (-4)$

• It can't be done as easily mentally,

• Students see it as "yet another formula to learn" when really all you need is to appreciate is what gradient is, i.e. "y change per x change".

Chapter 5 · Original slide 9 · Examples

Find the gradient of the line that goes through the points:

1

$(1, 4)(3, 10)$

2

$(5, 7)(8, 1)$

3

$(2, 2)(-1, 10)$

4

Show that the points $\mathrm{A}(3, 4), B(5, 5), C(11, 8)$ all lie on a straight line.

If points are 'collinear' they lie on the same line.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 10 · Further Example

The line joining $(2, -5)$ to $(4, a)$ has gradient -1.
Work out the value of a.

Chapter 5 · Original slide 11 · y=mx+c

One form we can put a straight line equation in is:

$y = mx + c$

Gradient

y-intercept

Why does it work?

• The $\mathbf{y}$-intercept by definition is the $\mathbf{y}$ value when $\mathbf{x=0}$. Substituting:
$\mathbf{y=(m\times 0)+c=c}$ as expected.

• By the definition of gradient, if we increase $\mathbf{x}$ by 1, $\mathbf{y}$ should increase by $\mathbf{m}$:
$\mathbf{y=m\,(x+1)+c=mx+m+c}$
which indeed has increased by $\mathbf{m}$.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 12 · y=mx+c

Determine the gradient and y-intercept of the line with equation $4x - 3y + 5 = 0$

Make y the subject so we have the form

$y = mx + c$

Put y on the side it's positive.

Divide each term by 3; don't write $y = \frac{4x+5}{3}$ otherwise it's not in the form $y = mx + c$

This is algebra, so use improper fractions, and not mixed numbers or recurring decimals.

Chapter 5 · Original slide 13 · ax+by+c=0

At GCSE, $\mathbf{y=mx+c}$ was the main form you would express a straight line equation, sometimes known as the 'slope-intercept form'.

But another common form is $\mathbf{ax+by+c=0}$, where a, b, c are integers. This is known as the 'standard' form.

Express $y = \frac{1}{3}x - \frac{2}{3}$ in the form $ax + by + c = 0$, where a, b, c are integers.

We don't want fractions, so multiply by an appropriate number.

Put everything on either side of equation. $-x + 3y + 2 = 0$ would also be OK.

We'll see on the next slide WHY we might want to put an equation in this form over $y = mx + c$...

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 14 · Just for your interest…

Why might we want to put a straight line equation in the form $ax + by + c = 0$?

$2x + y = 4$



$y = mx + c$

"Slope-Intercept Form"

$ax + by + c = 0$

"Standard Form"

Coverage

$y = mx + c$ doesn't allow you to represent vertical lines. Standard form allows us to do this by just making b zero.

Usefulness

This more 'elegant' form also means it ties in with vectors and matrices. In FM, you will learn about the 'dot product' of two vectors:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = ax + by$$

thus since $ax + by + c = 0$, we can represent a straight line using:

$$\begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} + c = 0 (1)$$

Symmetry

In general, the 'linear combination' of two variables x and y is $ax + by$, i.e. "some amount of x and some amount of y". There is a greater elegance and symmetry to this form over $y = mx + c$ because x and y appear similarly within the expression.

We can extend to 3D points to get the equation of a plane:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \end{pmatrix} + d = 0 (2)$$

Conveniently, in equation (1), the vector $\begin{pmatrix} a \\ b \end{pmatrix}$ is perpendicular to the line. And in equation (2), the vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$ is perpendicular to the plane. Nice!

$x + 4 = 0$

$$\begin{pmatrix} 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = 4$$

$\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

Chapter 5 · Original slide 15 · Test Your Understanding

Express $y = \frac{2}{5}x + \frac{3}{5}$ in the form $ax + by + c = 0$, where a, b, c are integers.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 16 · Exercise 5A – Pure Y1 Page 91

10 Show that the points $A(-2a, 5a)$, $B(0, 4a)$ and points $C(6a, a)$ are collinear. (3 marks)

9 Show that the points $A(2, 3)$, $B(4, 4)$ and $C(10, 7)$ can be joined by a straight line.

Chapter 5 · Original slide 17 · Exercise 5B – Pure Y1 Page 92-93

8 A line l cuts the x-axis at $(5, 0)$ and the y-axis at $(0, 2)$.

a Find the gradient of the line.

b Find an equation of the line in the form $ax + by + c = 0$.

10 The line l with gradient 3 and y-intercept $(0, 5)$ has the equation $ax - 2y + c = 0$.

Find the values of a and c. (2 marks)

9 Show that the line with equation $ax + by + c = 0$ has gradient $-\frac{a}{b}$ and cuts the y-axis at $-\frac{c}{b}$.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 18 · Exercise 5B – Pure Y1 Page 92-93

11 The straight line l passes through $(0, 6)$ and has gradient -2. It intersects the line with equation $5x - 8y - 15 = 0$ at point P. Find the coordinates of P.

12 The straight line l_1 with equation $y = 3x - 7$ intersects the straight line l_2 with equation $ax + 4y - 17 = 0$ at the point $P(-3, b)$.

a Find the value of b.

b Find the value of a.

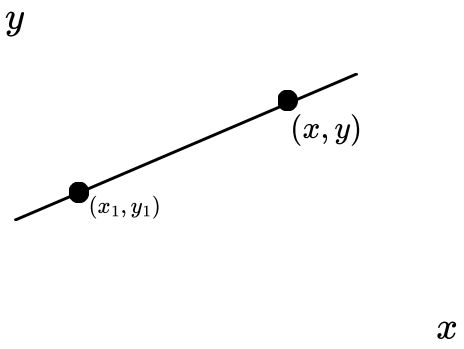
Chapter 5 · Original slide 19 · Equations using two points/point + gradient

Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

How would you have done this at GCSE?

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 20 · A new way…



Suppose that (x_1, y_1) is some fixed point on the line that we specify (e.g. $(3, 5)$).

Suppose that (x, y) represents a generic point on the line, which is allowed to change as we consider different points on this line.

Then:

$m = \frac{y-y_1}{x-x_1}$

Thus:

Notes: Note that x_1 and y_1 are constants while x and y are variables. The latter are variables because as these 'vary', we get different points on the line.

! The equation of a line that has gradient m and passes through a point (x_1, y_1) is:

$y - y_1 = m(x - x_1)$

Let's revisit:

Find the equation of the line that goes through $(3, 5)$ and has gradient 2.

No manipulation required! Just plug in numbers and we're done.

Again, note that x and y are variables so need to remain as such.

Chapter 5 · Original slide 21 · Quickfire Questions

In a nutshell: You can use this formula whenever you have (a) a gradient and (b) any point on the line.

Gradient	Point	(Unsimplified) Equation
3	$(1, 2)$	
5	$(3, 0)$	
2	$(-3, 4)$	
$\frac{1}{2}$	$(1, -5)$	
9	$(-4, -4)$	

Important Side Note: I've found that many students shun this formula and just use the GCSE method. Please persist with it – it'll be much easier when fractions are involved.

Further Mathematicians, don't even think about using the GCSE method, because you'll encounter massive headaches when you consider algebraic points. Trust me on this one!

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 22 · Using 2 points

Find the equation of the line that goes through $(4, 5)$ and $(6, 2)$, giving your equation in the form $ax + by + c = 0$.

We need to work out the gradient first before we can use $y - y_1 = m(x - x_1)$:

Test Your Understanding:

Find the equation of the line that goes through $(-1, 9)$ and $(4, 5)$, giving your equation in the form $ax + by + c = 0$.

Chapter 5 · Original slide 23 · Exercise 5C – Pure Y1 Page 94-95

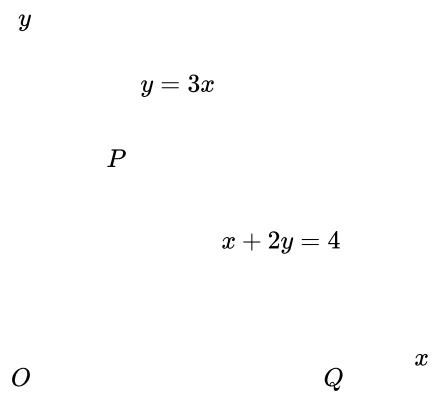
3 Find the equation of the line l which passes through the points $A(7, 2)$ and $B(9, -8)$. Give your answer in the form $ax + by + c = 0$.

6 The straight line l passes through $(7a, 5)$ and $(3a, 3)$. An equation of l is $x + by - 12 = 0$. Find the value of a and the value of b.

5 The straight line l passes through $(a, 4)$ and $(3a, 3)$. An equation of l is $x + 6y + c = 0$. Find the value of a and the value of c.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 24 · Intersection of lines



The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

We know from Chapter 4 we can just solve two equations simultaneously.

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the coordinate of Q.

Chapter 5 · Original slide 25 · Test Your Understanding

The straight line L_1 passes through the points $(-1, 3)$ and $(11, 12)$.
(a) Find an equation for L_1 in the form $ax + by + c = 0$, where a, b and c are integers. (4)
The line L_2 has equation $3y + 4x - 30 = 0$.
(b) Find the coordinates of the point of intersection of L_1 and L_2. (3)

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1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 26 · Exercise 5D – Pure Y1 Page 96

6 The line that passes through the points $(-3, -5)$ and $(4, 9)$ meets the y-axis at the point G. Work out the coordinates of the point G.

9 The lines $y = 4x - 10$ and $y = x - 1$ intersect at the point T. Find the equation of the line with gradient $-\frac{2}{3}$ that passes through the point T. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

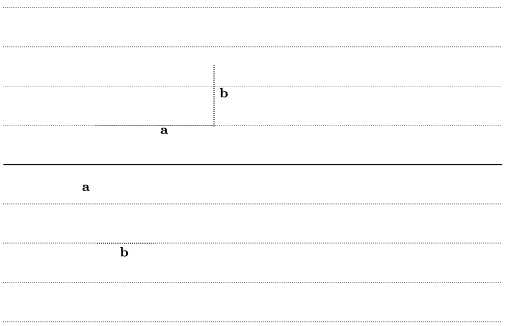
Chapter 5 · Original slide 27 · Exercise 5D – Pure Y1 Page 96

14 The line $y = 4x + 5$ meets the y-axis at the point C. The line $y = -3x - 15$ meets the x-axis at the point D. Find the equation of the line joining the points C and D. Write your answer in the form $ax + by + c = 0$, where a, b and c are integers.

16 The lines $y = -2x + 1$ and $y = x + 7$ intersect at the point L. The point M has coordinates $(-3, 1)$. Find the equation of the line that passes through the points L and M.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 28 · y



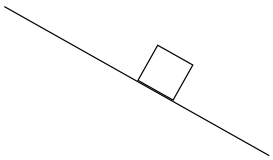
4
3
2
1
-1
-2
-3
-4

x -5 -4 -3 -2 -1 0 1 2 3 4 5 6

Find the gradients of each pair of perpendicular lines. What do you notice?

Using the changes above, we can see the gradient of one line is $\mathbf{m=\frac{b}{a}}$ and the other $\mathbf{m=-\frac{a}{b}}$. One is the 'negative reciprocal' of the other.

Chapter 5 · Original slide 29 · Perpendicular Lines



The gradients of parallel lines are equal.

If two lines are perpendicular, then the gradient of one is the negative reciprocal of the other.

$m_1 = -\frac{1}{m_2}$

To show that two lines are perpendicular:

$m_1 m_2 = -1$

Gradient	Gradient of Perpendicular Line
2	
-3	
$\frac{1}{4}$	
5	
$-\frac{2}{7}$	
$\frac{7}{5}$	

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 30 · Example Problems

1 2 3

A line is goes through the point (9,10) and is perpendicular to another line with equation $y = 3x + 2$. What is the equation of the line?

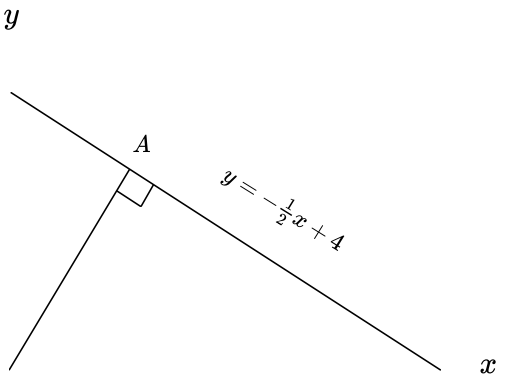
A line L_1 goes through the points $A(1, 3)$ and $B(3, -1)$. A second line L_2 is perpendicular to L_1 and passes through point B. Where does L_2 cross the x-axis?

Are the following lines parallel, perpendicular, or neither?

$y = \frac{1}{2}x$

$2x - y + 4 = 0$

Chapter 5 · Original slide 31 · Test Your Understanding



2

Determine the point A.

1

A line goes through the point (4,7) and is perpendicular to another line with equation y=2x+2. What is the equation of the line? Put your answer in the form ax+by+c=0, where a,b,c are integers.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 32 · Exercise 5E/5F – Pure Y1 Page 96

3 The coordinates of a quadrilateral $ABCD$ are $A(-6, 2), B(4, 8), C(6, 1)$ and $D(-9, -8)$. Show that the quadrilateral is a trapezium.

7 A line is parallel to the line $2x - 3y - 1 = 0$ and it passes through the point $(0, 0)$. Write down the equation of the line.

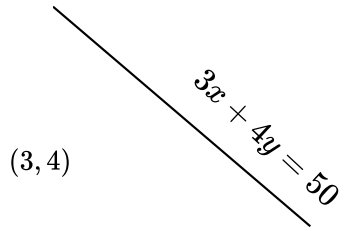
Chapter 5 · Original slide 33 · Exercise 5F – Pure Y1 Page 100

9 The line l passes through the points $(-3, 0)$ and $(3, -2)$ and the line n passes through the points $(1, 8)$ and $(-1, 2)$. Show that the lines l and n are perpendicular.

10 The vertices of a quadrilateral $ABCD$ have coordinates $A(-1, 5), B(7, 1), C(5, -3)$ and $D(-3, 1)$. Show that the quadrilateral is a rectangle.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 34 · Exercise 5E/5F – Pure Y1 Page 96



Extension Problems

2

[MAT 2014 1D] The reflection of the point $(1, 0)$ in the line $y = mx$ has coordinates: (in terms of m)

1

[MAT 2004 1D]

What is the reflection of the point $(3, 4)$ in the line $3x + 4y = 50$?

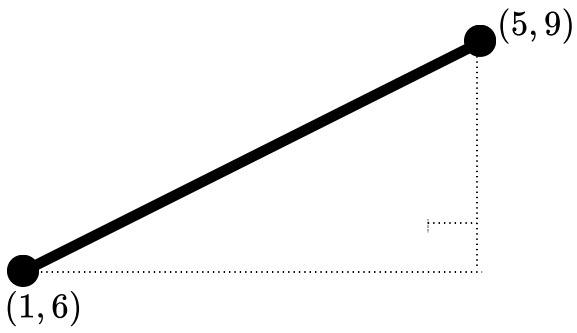
3

[STEP I 2004 Q6] The three points A, B, C have coordinates $(p_1, q_1), (p_2, q_2)$ and (p_3, q_3), respectively. Find the point of intersection of the line joining A to the midpoint of BC, and the line joining B to the midpoint of AC. Verify that this point lies on the line joining C to the midpoint of AB.

The point H has coordinates $(p_1 + p_2 + p_3, q_1 + q_2 + q_3)$. Show that if the line AH intersects the line BC at right angles, then $p_2^2 + q_2^2 = p_3^2 + q_3^2$, and write down a similar result if the line BH intersects the line AC at right angles.

Deduce that if AH is perpendicular to BC and also BH is perpendicular to AC, then CH is perpendicular to AB.

Chapter 5 · Original slide 35 · Distances between points



Recall: Δ (said 'delta') means "change in".

How could we find the distance between these two points?

Hint: Pythagoras

Form a right-angled triangle using the change in x and change in y, then use Pythagoras.

! Distance between two points:

$$\sqrt{(\Delta x)^2 + (\Delta y)^2}$$

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 36 · Examples

Distance between:

(3,4) and $(5, 7)$

(5,1) and $(6, -3)$

(0,-2) and $(-1, 3)$

Note: Unlike with gradient, we don't care if the difference is positive or negative (it's being squared to make it positive anyway!)

Quickfire Questions:

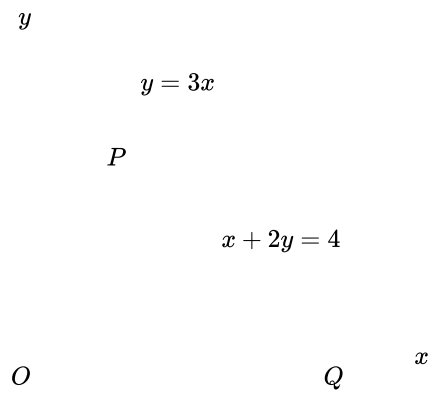
Distance between:

(1,10) and $(4, 14)$

(3,-1) and $(0, 1)$

(-4,-2) and $(-12, 4)$

Chapter 5 · Original slide 37 · Area of Shapes



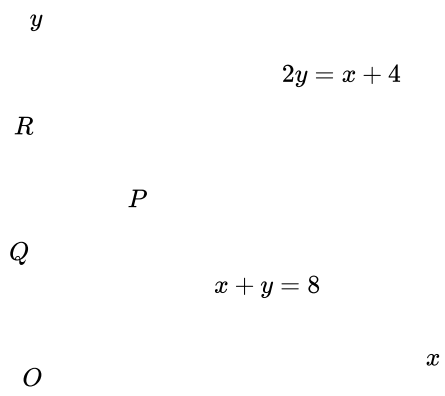
The diagram shows two lines with equations $y = 3x$ and $x + 2y = 4$, which intersect at the point P.

a) Determine the coordinates of P.

(We did this in a previous lesson)

b) The line $x + 2y = 4$ intersects the x-axis at the point Q. Determine the area of the triangle OPQ.

Chapter 5 · Original slide 38 · Further Example



Tip: When finding areas of triangles in exam questions, one line is often vertical or horizontal. You should generally choose this to be the 'base' of your triangle.

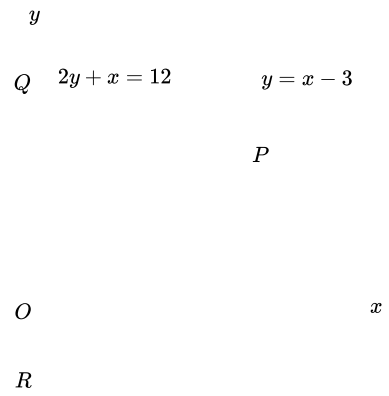
a

Determine the length of PQ.

b

Determine the area PQR.

Chapter 5 · Original slide 39 · Test Your Understanding



a) Determine the coordinate of P.

b) Determine the area of PQR.

c) Determine the length PQ.

Chapter 5 · Original slide 40 · Exercise 5G – Pure Y1 Page 102

7 A point P lies on the line with equation $y = 4 - 3x$. The point P is a distance $\sqrt{34}$ from the origin. Find the two possible positions of point P. (5 marks)

8 The vertices of a triangle are $A(2, 7)$, $B(5, -6)$ and $C(8, -6)$.
a Show that the triangle is a scalene triangle.
b Find the area of the triangle ABC.

Chapter 5 · Original slide 41 · Exercise 5G – Pure Y1 Page 103

12 The straight line l_1 passes through the point $(-4, 14)$ and has gradient $-\frac{1}{4}$.
a Find an equation for l_1 in the form $ax + by + c = 0$, where a, b and c are integers.
b Write down the coordinates of A, the point where straight line l_1 crosses the y-axis.
The straight line l_2 passes through the origin and has gradient 3. The lines l_1 and l_2 intersect at the point B.
c Calculate the coordinates of B.
d Calculate the exact area of $\triangle OAB$.

1.03 Coordinate Geometry: Linear Graphs

Chapter 5 · Original slide 42 · Exercise 5G – Pure Y1 Page 102-103

Extension Problems

1

[MAT 2001 1C]

The shortest distance from the origin to the line $3x + 4y = 25$ is what?

$3x + 4y = 25$

$(0, 0)$